

50. A 45 year old female patient admitted to Tikur Anbessa Hospital has an alveolar ventilation of 5L/min, frequency of breathing 10 breath/min, and tidal volume of 700ml. Her vital capacity is 4L. What is her dead space ventilation?

- A. 2.5 L/min
- B. 3.5L/min
- ☒ C. 2L/min
- D. 0.5L/min
- E. 1L/min

51. Gas exchange at the tissue bed and pulmonary bed involves

- A. Active transport mechanism
- B. Facilitated transport mechanism
- ☒ C. Simple diffusion
- D. Against partial pressure gradient
- E. Carrier mediated transport

52. Which ONE of the following metabolic conditions occurs in ischemic conditions?

- A. Increase in FA oxidation
- B. Increase in glucose oxidation
- ☒ C. Inhibition of PDH activity by increased level of acetyl-coA
- D. Decrease in lactate accumulation
- E. Switch to aerobic metabolism

53. Which of the following glycolytic enzyme is a rate limiting?

- ☒ A. Phosphofructokinase-1
- B. Glyceraldehyde-3-phosphate dehydrogenase
- C. Pyruvate kinase
- D. Glucokinase
- E. Phosphofructokinase-2

54. Which one of the following particles is harming the Endothelial Cells?

- A. High LDL-C
- B. High HDL-C
- ☒ C. High Glucose
- D. High Free fatty acid
- E. High Amino acids

55. Activated inflammasomes increase lipid synthesis but decrease lipid oxidation because the lipids are needed for:

- ☒ A. Plasma membrane repair
- B. Adipogenesis
- C. Bile synthesis
- D. Angiogenesis
- E. Insulation

56. Which one of the following molecules promotes atherosclerosis?

- A. Apo B 100
- B. Apo A-I
- C. Apo A-II
- D. Apo C-II
- E. Apo C-III

57. Why Statins are not considered as best therapeutic agents for atherosclerosis?

- A. They reduce the amount of LDL-C but not improve the LDL-C size
- B. They reduce the LDL-C but not improve the HDL-C levels
- C. They don't reduce the LDL-C but improve the HDL-C levels
- D. They reduce the LDL-C but not the levels of triacylglycerols
- E. They reduce both LDL-C and HDL-C levels

58. What type of lipoprotein changes occur during metabolic syndrome?

- A. More Apo A-I and less Apo B 100
- B. Less Apo A-I and less Apo B 100
- C. Less Apo A-I and more Apo B 100
- D. More Apo A-I and more Apo B 100
- E. Normal Apo A-I and more Apo B 100

59. A 13-year-old boy complained of headache, muscle pains, fever weight loss for 2 months. On examination his temperature was 38.0°C. An apical systolic heart murmur was noted. Blood cultures grew viridans streptococci susceptible to penicillin. The next day the patient underwent mitral valve repair and was discharged 5 days later. The most likely disease is:

- A. Pericarditis
- B. Myocarditis
- C. Subacute bacterial endocarditis
- D. Rheumatic heart disease
- E. Endarteritis obliterans

60. Which of the following viral pathogens is the main cause of myocarditis?

- A. Influenza viruses
- B. Hepatitis viruses
- C. Coxsackie viruses
- D. Rhinoviruses
- E. Adenoviruses

61. Viridans streptococci are:

- A. Obligate pathogens
- B. Normal oral commensals
- C. Beta-hemolytic
- D. Gamma-hemolytic
- E. Optochin positive

62. The most important virulence factor for *Streptococcus pneumoniae* is

- A. Toxin
- B. Enzyme
- C. Capsule
- D. Flagella
- E. Pili

63. The action of streptolysin results in

- A. The breaking of the hyaluronic acid capsule around cells
- B. The inhibition of complement protein
- C. The increasing in the number of leukocytes at the site of infection
- D. Breaks in the membranes of erythrocytes, leukocytes, and platelets
- E. The destruction of streptococcal bacteria

64. The mechanism of action of diphtheria toxin is to

- A. disrupt the cell membrane
- B. block nucleic acid synthesis
- C. block protein synthesis
- D. interfere with neurotransmission
- E. destroy the cell nucleus

65. Dysphonia (changes in the voice) in a 2-year-old child with a grayish exudate with a pseudomembrane in the posterior pharynx and massive, tender cervical lymphadenitis is most likely caused by

- A. *Corynebacterium diphtheriae*
- B. *Neisseria meningitidis*
- C. *Streptococcus pyogenes*
- D. Rhinoviruses
- E. *Bordetella pertussis*

66. Which respiratory infection-associated viruses are identified into different serotypes with their hemagglutinin and neuraminidase combinations such as H1N1, H2N1, H1N2, etc.?

- A. Respiratory Syncytial virus
- B. Influenza virus type A
- C. Influenza virus type 2
- D. Influenza virus type C
- E. Parainfluenza virus

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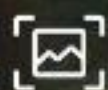
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- all of the following are causes for flu virus pathogenesis EXCEPT:
- A. Destruction of epithelial cells that leads to loss of mucociliary flow, which in turn hinders removal of foreign bodies including bacteria and virus, giving them sufficient exposure time to initiate infection.
 - B. Production of high amount of Interferon as a defense response to influenza virus infection, which leads to strong local and systemic inflammations that result in "Cytokine storm" which cause severe clinical manifestations.
 - C. Poor and incompetent immune status of the infected person may fail to clear the viruses, as the result of which the virus may disseminate to other anatomic sites where they cause pneumonia, central nervous system infection, and infection involving muscle and heart.
 - D. Antibody mediated immune response (humeral immunity) against influenza virus infection almost always produces non-neutralizing antibodies that bind with the virus without neutralizing them and even enhance the antibody-mediated receptor attachment to Fc bearing target cells.
 - E. The cell-mediated immune response, in addition to triggering system inflammation, is also responsible for direct killing of the infected cells using cytotoxic T lymphocytes and killing by inducing apoptosis.
68. Which of the following is not part of differential diagnosis of Influenza A infection?
- A. Infection by Adenovirus
 - B. Infection by parainfluenza virus
 - C. Infection by Rabies virus
 - D. Infection by human coronavirus
 - E. Infection by Mycoplasma pneumoniae
69. Active immunization against flu _____:
- A. makes use of vaccine developed from strains that caused the 2018
 - B. can be prepared from killed whole virus which are effective in elderly and children not in young adults
 - C. is currently prepared from HA subunits from recombinant
 - D. from live attenuated viruses have never been prepared during each season.
 - E. is manufactured by pharmaceutical industries by combining drugs such as Rimantadine and Oseltamivir
70. Histoplasma capsulatum produces _____:
- A. Intra and extracellular yeast cells at 25°C
 - B. only macroconidia at 25°C
 - C. tuberculate chlamydospores at 37°C
 - D. does not grow at 25°C
 - E. tuberculate chlamydospores at 25°C

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- A. makes use of vaccine developed from strains that caused the 2018 Spanish flu
- B. can be prepared from killed whole virus which are effective largely in the elderly and children not in young adults
- C. is currently prepared from HA subunits from recombinant viruses circulating during each season.
- D. from live attenuated viruses have never been prepared and used in any country because these viruses can very easily reverse to infectious form.
- E. is manufactured by pharmaceutical industries by combining Chemoprophylaxis drugs such as Rimantadine and Oseltamivir

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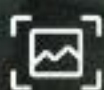
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Choose the best answer from the given alternatives in each question.

1. Which of the following statements is NOT CORRECT with regard to the *circulatory system*?
- A. Lymphatic capillaries can penetrate the media of veins.
 - B. Vasa vasorum are more frequent in veins than in arteries.
 - C. Post-capillary venules are sites of exit of erythrocytes from blood into tissue by diapedesis.
 - D. The tunica adventitia tends to be the thickest and the most prominent in veins than in arteries.
 - E. Pericytes are contractile mesenchymal cells that participate in repair processes.
2. Select the statements which is NOT CORRECT with regard to the *histology of the heart*.
- A. Pectinate muscles are the sites of attachment of chordae tendinae.
 - B. Purkinje cells travel in subendocardial layer of the endocardium to the apex of the heart.
 - C. Epicardium is a serous covering of the heart, forming the visceral layer of the pericardium.
 - D. Endocardium is the thickest in atria and the thinnest in ventricles, especially the left ventricle.
 - E. Cardiac muscle is more vascularized and has more mitochondria than does skeletal muscle.
3. Which of the following components of the *respiratory tract* wall increase(s) in amount from trachea to alveoli?
- A. Cilia
 - B. elastic fibers
 - C. smooth muscle
 - D. cartilage
 - E. goblet cells
4. Which of the following statements is CORRECT regarding the *respiratory system*?
- A. There are *seromucous glands* and *venous plexuses* in the alveoli.
 - B. Pulmonary surfactant prevents alveolar collapse by increasing surface tension.
 - C. The superior conchae are covered with stratified squamous nonkeratinized epithelium.
 - D. The lining of the pleural cavity is made up of pseudostratified ciliated columnar epithelium.
 - E. Smooth muscle is present in the entire bronchiolar tree, including the *respiratory bronchiole*.

- Concerning the development of the heart, which statement is CORRECT?
- A. The truncus arteriosus forms the roots and proximal portion of the aorta and pulmonary artery.
 - B. Sinusatrial nodes are derived from cells of the atrioventricular canal during the 8th week.
 - C. The venous end of the heart during the 3rd week is fixed by the septum transversum.
 - D. The left horn of sinus venosus is incorporated into the wall of the right atrium.
 - E. The bulbus cordis forms the trabeculated part of the right ventricle.
6. Which of the following is NOT associated with Tetralogy of Fallot?
- A. Pulmonary stenosis
 - B. Ventricular septal defect
 - C. Pulmonary atresia
 - D. Right ventricular hypertrophy
 - E. Dextroposition of the Aorta (over riding aorta)
7. Which of the following cells of the lung develop from mesoderm?
- A. Type I Pneumocytes.
 - B. Clara cells
 - C. Type II Pneumocytes
 - D. Alveolar dust cells
 - E. Goblet cells
8. Select the incorrect statement with regard to the development of the respiratory system.
- A. Surfactant production begins by week 20.
 - B. Sacs analogous to alveoli are present at 32 weeks.
 - C. Cartilages of the larynx develop from the 4th and 6th pairs of pharyngeal arches.
 - D. Epiglottis develops by mesenchymal proliferation of the 2nd pharyngeal arches.
 - E. Embryonic right main bronchus is slightly larger than the left and is oriented more vertically.
9. Which one of the following is not a feature of right atrium?
- A. Cristae terminalis
 - B. Fossa ovalis
 - C. opening of coronary sinus
 - D. ☒ None
10. The apex of the heart lies in
- A. ☒ left 5th intercostal space just medial to midclavicular line
 - B. left 2nd intercostal space just lateral to sternum
 - C. right 2nd intercostal space just lateral to sternum
 - D. right 5th intercostal space just medial to midclavicular line
11. What structures work to prevent prolapse (eversion) of the atrioventricular valves?
- A. Trabeculae carneae
 - B. ☒ Papillary muscles
 - C. Pectinate muscles
 - D. Autorhythmic cells



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12. One of the following is not functions of the lymphatic system?
A. To return water and proteins from the interstitial fluid to the blood.
B. To transport fat absorbed from the GIT to the blood
C. Haemopoiesis
D. None
13. Costal cartilage of one of the following rib is articulated at sternal angle
A. Rib 4
B. Rib 2
C. Rib 1
D. Rib 3
14. One of the following event does not takes place behind the sternal angle?
A. pulmonary trunk divides
B. trachea bifurcates
C. arch of the aorta begins and ends
D. SVC formed
15. Some of the thoracic ribs are classified under typical ribs, because
A. They are attached with vertebral column by their own costal cartilage
B. They are attached with sternum by their own costal cartilage
C. They are supplied by their independent arteries
D. They are located on the top of thoracic cage
E. all
16. One of the following structure pass through aortic opening?
A. Thoracic duct
B. Vagus nerve
C. Left phrenic nerve
D. Splanchnic nerves
17. One of the following statement is not true regardless of azygous veins system?
A. The 1st posterior intercostal vein drains in to respective brachiocephalic vein
B. Posterior superior intercostal vein drains in to brachiocephalic vein on both sides
C. Azygous vein joins the superior venacava
D. None



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18. One of the following statement is not true regardless of the general feature of the lungs?
- A. heavier in the male than in the female
 - B. occupies most of the thoracic cavity
 - C. The apex directed inferiorly
 - D. None
19. One of the following is false about the hilar structures of the lungs?
- A. The pulmonary artery is the most superior organ on the left lung
 - B. The pulmonary vein is the most superior organ on the right lung
 - C. The bronchus is the most posterior organ on both lungs
 - D. The pulmonary vein is most anterior organ on both lungs
 - E. The pulmonary vein is the most inferior organ on the left lung
20. Accumulation of air with in thoracic cavity is termed as
- A. Pneumothorax
 - B. Hydrothorax
 - C. Chylothorax
 - D. Hemothorax
21. A -66 year old man experiences orthostatic hypotension following sympathectomy. The explanation to this occurrence is:
- A. An exaggerated response of the renin-angiotensin -aldosterone system (RAAS)
 - B. A suppressed response of the RAAS system
 - C. An exaggerated response of the baroreceptor reflex mechanism
 - D. A suppressed response of the baroreceptor reflex response
 - E. An exaggerated shift of blood flow to the kidneys
22. The sinoatrial nodal cells (SA node) are:
- A. Innervated by somatic nerve cells
 - B. Found in both atria
 - C. Able to generate impulses because their membrane potential is unstable
 - D. Connected to the atrioventricular node by fine bundles of Purkinje cells
 - E. Unable to generate impulses when completely denervated
23. Which of the following statements is not true:
- A. Ventricular muscle is electrically refractory during the period of rapid ejection
 - B. The tricuspid and mitral valves open at the same time
 - C. Ejection begins when the ventricular pressure exceeds 80mmHg in both the right and left ventricles
 - D. The T wave of the ECG occurs during the ejection phase of the cardiac cycle
 - E. Atrial contraction contributes only 30% of ventricular filling in a resting person
24. Second heart sound occurs at the onset of which phase of the cardiac cycle?
- A. isovolumetric contraction
 - B. rapid ejection
 - C. isovolumetric relaxation
 - D. atrial systole
 - E. rapid ventricular filling

25. Which of the following conditions would increase the afterload placed on the ventricle?
- A. Hypertension
 - B. Aortic valve stenosis
 - C. Increased total peripheral resistance
 - D. Emboli
 - E. All of the above
26. Blood flow through a vessel varies directly with
- A. Resistance
 - B. Viscosity
 - C. Pressure gradient
 - D. Length of blood vessel
 - E. All are correct
27. Blood vessels that contribute most to blood pressure regulation is
- A. Aorta
 - B. Large arteries
 - C. Small arteries
 - D. Capillaries
 - E. Veins
28. Cardiac muscle differs from skeletal muscle by their difference
- A. In the sensitivity to ECF calcium
 - B. Timing of opening of K channel
 - C. Type of action potential
 - D. In response to Adrenaline
 - E. All are correct
29. Systolic murmur may be due to
- A. AV valve stenosis
 - B. Aortic valve stenosis
 - C. Aortic valve insufficiency
 - D. Ventricular hypertrophy
 - E. Anemia
30. The first line and rapid ABP regulatory system that prevent postural hypotention is
- A. Baroreceptor reflex
 - B. Chemoreceptor reflex
 - C. Cardiopulmonary reflex
 - D. Renal body fluid mechanism
 - E. Fluid shift from IF to plasma
31. Windkessel function of our circulatory system played by is
- A. Large arteries
 - B. Smaller arteries
 - C. Capillaries
 - D. Veins
 - E. Lymphatics

32. Veins are
- A. Exchange areas
 - B. Serve stop cock function
 - C. Windkessels
 - D. Blood reservoirs
 - E. High high pressure are
33. Aortic regurgitation is prevented by
- A. Early closure of aortic valve
 - B. Late closure of aortic valve
 - C. Low pressure of left ventricles
 - D. Closure of AV valve
 - E. Opening of AV valve
34. Normal functioning of the heart require all of the following except
- A. All valves must open and close simultaneously
 - B. Contraction must occur at regular interval
 - C. Muscle contraction must be happen
 - D. Ventricle must fill adequately
 - E. Valves must open fully
35. Which of the following factors influence cardiac muscle activity by altering the level of cAMP?
- A. Thyroid hormone
 - B. Digitalis
 - C. Calcium
 - D. Catecholamines
 - E. Hypertrophy
36. Which one is NOT correct regarding ventilation during low-medium work intensity?
- A. Ventilation increases linearly with intensity
 - B. Maintains arterial acid-base status
 - C. CO_2 is removed in exact proportion to CO_2 produced.
 - D. Lactic acidosis-induces hyperventilation
 - E. Compensation is primarily by increasing the depth of breathing
37. During exercise, increase in O_2 uptake by exercising muscles occurs due to all of the following mechanisms except:
- A. oxyhemoglobin dissociation curve shifts to the left
 - B. decrease in vascular resistance in exercising skeletal muscle
 - C. Increased O_2 extraction
 - D. Increased blood flow
 - E. Increased 2,3-diphosphoglycerate

38. Which of the following is a cardiovascular or aerobic activity that stimulates the use of fat?
- A. Jumping
 - B. Weight lifting
 - C. Sprinting (100 meter)
 - D. Jogging
 - E. Boxing
39. Which of the following one is not part of the biochemical and physiological responses of humans to hypoxia at high altitude (~ 4,000 m):
- A. Is associated with the development of respiratory alkalosis
 - B. Results in elevation of alveolar PO₂ relative to inspired PO₂.
 - C. **Is the result of increased sensitivity of central chemoreceptors to oxygen**
 - D. Results in improved unloading of oxygen at the tissues but an impaired uptake of oxygen at the lung
 - E. Results in an increased capacity of the animal to metabolize aerobically.
40. Ethiopian way of adapting to high altitude which is said to be the third successful pattern is characterized by
- A. Erythrocytosis without arterial hypoxemia
 - B. **Absence of both marked erythrocytosis and arterial hypoxemia**
 - C. Normal venous hemoglobin with arterial hypoxemia
 - D. Presence of both erythrocytosis with arterial hypoxemia
 - E. Marked hyperventillatory response
41. All of the following are involved in non-breathing functions of the respiratory system **except**:
- A. Defense against viruses
 - B. pH regulation
 - C. Antigen presentation
 - D. Ventilation
 - E. Electrolyte regulation
42. Why nasal breathing is advantageous over mouth breathing?
- A. Nasal breathing increases surface area for gas exchange
 - B. Mouth breathing causes decreased reserve volumes
 - C. Nasal breathing cleanses the incoming air
 - D. Nasal breathing increases pulmonary compliance
 - E. Mouth breathing has increased work of breathing
43. In healthy individuals, the pulmonary and systemic circulations have almost the same:
- A. Mean vascular pressure
 - B. Vascular resistance
 - C. **Flow per minute**
 - D. Leakiness to plasma proteins
 - E. Blood volume

44. The respiratory unit includes all of the following EXCEPT:

- A. Respiratory bronchioles
- B. Alveolar duct
- C. Alveolar duct
- D. Terminal bronchioles
- E. None of the above

45. Identify the correct statement about autonomic innervations of the respiratory system:

- A. Parasympathetic decreases air way resistance work of breathing
- B. Sympathetic nerve supply arises from sacral segments of spinal cord
- C. Sympathetic activation causes bronchodilation
- D. Autonomic activation controls the smooth muscles of alveoli
- E. Three of the above choices are correct

46. Some premature newborns lack pulmonary surfactant; therefore,

- A. Their lungs are difficult to re-inflate
- B. Their work of breathing is reduced due to decreased surface tension
- C. They have impaired function of type I alveolar epithelial cells
- D. Their lung compliance is increased
- E. Their respiratory muscles atrophy

47. Which of the following best characterizes blood flow, and alveolar ventilation-ventilation-perfusion ratio at the base, compared with the apex, of the lungs of a healthy standing person?

Blood flow	Air flow	Ventilation-perfusion ratio
A. Lower	Higher	Higher
B. Higher	Higher	Lower
C. Lower	Lower	lower
D. Higher	Lower	Higher
E. Lower	Lower	Higher

48. Gas exchange at the tissue bed and pulmonary bed involves

- A. Active transport mechanism
- B. Facilitated transport mechanism
- C. Simple diffusion
- D. Against partial pressure gradient
- E. Carrier mediated transport

49. In asthmatic patients, as compared to normal people which type of work done by the breathing muscles is increased remarkably?

- A. Compliance work
- B. Tissue resistance work
- C. Elastic work
- D. Airways resistance work
- E. Inelastic work